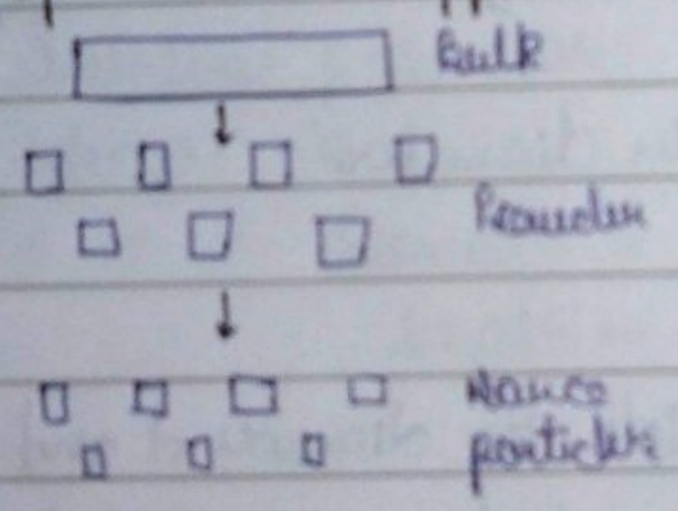
* UNIT - 2 => NANOMATERIALS *

→ Anything which on the scale less than 100 nm.

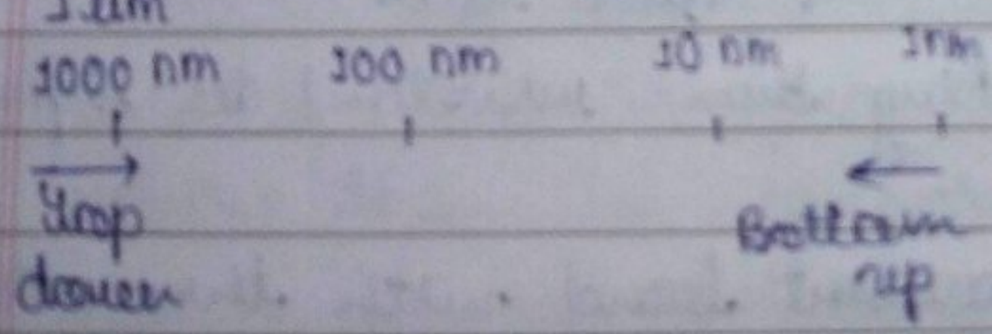
	Macroscale	Nano Scale
copper	opaque	transparent
Platinum	Inert	catalyst
Aluminium	Stable	combustible
Silica	Insulator	conductor
Gold	Solid Inert yellow	liq. catalyst. Red

* Method of synthesis

a) Top - down approach



→ High energy wet ball milling
→ electron beam lithography
→ Atomic force manipulation
→ Gas phase condensation
→ Thermal spray.



b) Bottom - up approach

Bottom up

→ Organometallic chemical routes
→ Sol - Gel synthesis

* Carbon Nanotubes (CNT) ⇒
→ CNT are hollow tubes composed of C-atom arranged in hexagonal lattice.
→ Typically measuring 1-3nm in diameter.

• Properties:
→ More hardness as comparable to diamond and strength 10 times more than steel.
→ Have gravitational wt. 1/4th lower steel.
→ High heat capacity, making them resistant to expansion when heated.
→ Each C-atom form covalent bond with three others creating a hexagonal lattice.
→ chemically inert and corrosion resistant

* Types of carbon nanotubes (CNT)
(i) Single walled CNT (SWCNT)
(ii) Multi walled carbon nanotubes (MWCNT)

* SWCNT ⇒
1) cylindrical formed from rolled graphene sheet with 1 dimensional shape.
2) Are Armchair and zig-zag determined.
3) Armchair SWCNT are conductive, other are semiconductor
4) Unique electronic properties are due to their specific lattice arrangment.
5) Exceptional strength flexibility and electrical conductivity makes valuable in various tech. application.

• Properties ⇒
→ Act as semiconductor
→ Band gap, range from 0 to 2ev.
→ Are used in circuit disassembly due to their specific electrical properties.
→ Shape is tiny
diameter → 1-2 nm
length → 1-2 μm

• Applications ⇒
→ SWCNT serves as multi-functional attributes enhancing diverse material properties at minimum conc. industry utilize them in composite coating, fibres, structural component and electrochemical device.
→ Also used as adhesive, lubricant, antistatic products, conductive film and cable etc.

(i) In composite materials →
→ SWCNT are finding way into industry like automotive, energy, automotive and aerospace. They are used to strengthen the composite material in wind turbine blade, boat hulls and aircraft parts, making them lighter and stronger.

(ii) In industrial coating →
→ SWCNT are added to primers and paints for vehical coating equipment and storage containers due to its durability and anti-static properties.

(iii) In car tyres →
→ CNT boosts tyre quality by strengthing resistance fuel efficiency and grip requiring only small amount and no change to production method. They improve electrical conductivity without extra fillers like carbon black. SWCNT are expected to be used in 30% of premium tyres and 10% for standard tyres.

(iv) In electro chemical
power sources →
→ used in rechargable batteries for electric vehicles, laptops and mobile phones because it boosts storage capacity and lifespan.
→ Lithium ion batteries and alkaline cell.

(v) Enhancing Rubber goods →
→ strength elasticity, absorption resistance and colour retention it covers 1/3rd portion of rubber industry

* Multiwalled carbon Nanotubes (MWCNTs) →
→ Are like rolled up scrolls made of several layers. They have both inner and outer diameter. It have good storage and more resistant to chemical reaction.

• Properties →
→ outer diameter = 2 - 20 nm
→ Inner diameter = 1 - 3 nm
→ Length = 5 - 6 um
→ Exhibit good tensile strength due to strong sp bond between C - atom.
→ when added to material like thermoset or thermoplastic, greatly increases its strength and storage capacity.
→ Exhibit outstanding chemical and thermal stability enduring the temp. exceeding 600°c.

• Application →
(i) In batteries:
→ Replacing graphite with MWCNT doubles the storage capacity of lithium ion battery which essential for portable electronics MWCNT based electrode are lighter thinner and ten times more conductive than amorphous carbon reducing batteries weight and power need specially in electronic vehical.
→ In super capacitor MWCNT increases power density enabling faster charging time for the device like cell phone and laptop.

(ii) In solar cell:
→ Solar cell with tall towers of MWCNT grown on iron coated silicon sheet absorb more light each tower contain

million of vertical alligned multiwalled CNT. leading to efficiency packed at 45 to 90 to sun making them ideal for space application without requiring mechanical orientation.

(iii) In transistor :-
→ MWCNT ~~play~~ can be used as transistor in modern integrated circuit offering a potential alternative to traditional silicon. Switches they may initially require very low temp and operate at the scale of individual electrons.

(iv) Nano electronics:-
→ MWCNT play a vital role in nano- electronic due to their high conductivity offering potential as semiconductor alternative to the silicon with conductivity influence by their structure.

(v) In enhancing flat panel display:-
→ MWCNT are used in flat panel display forming cathode stripe on glass plate and interacting with phosphorous coated Indium - Tin - Oxide (ITO) anode stripe to create pixels.

(vi) In energy production and storage Application →
→ MWCNT are essential for energy production and storage offering superior property for electrodes in batteries fuel cell and electro-chemical application compared to traditional carbonaceous material like carbon fiber and graphite.

* Applications of nanomaterial →
• In ~~electrons~~ → electronics :-
→ Electronics are crucial in our daily lives powering device like smartphone, computer and TV nanomaterials such as CNT and graphine are revolutionary material for electronics by offering better creation of thinner, lighter and more efficient electronic circuit and display. Nano-electronics focus on making electronic device smaller lighter and more powerfull by manipulating CNT material at the nano scale. Scientist can unlock new branch for electronics. This field hold promise for improving energy efficiency enhancing device performance from wearable tech. One significant appl^n for nano-electronics is in energy storage and energy generation nanomaterial enable the development of display, screen with lower power consumption and memory chip capable of storing vast amount of data in compact space wearable electronics such as smart watches and fitness tracker benifit from nanomaterial light weighted and energy efficient properties. Moreover nano-electronics contribute to advancement in green technology like solar cell and super ~~capacitor~~ capacitor offering more efficient power solution. Overall nanomaterials play a vital role in shaping the future of electronics and making the way for smaller, smarter and more energy efficient device.

• Uses of nanomaterials in sensor →
→ Sensors are vital tool for detecting physical, chemical or biological signals can be greatly enhanced by incorporating nano-materials their tiny structure sensors by boosting sensitivity, selectivity and speed

of response. For eg. nano scale particles and films improve sensors to detect even small quantities of gases toxin or bio molecule. These sensors can seamlessly integrated into wearable or implantable device, simplifying health monitoring moreover nano materials unlock novel sensors like nano photonic sensors which exploit light to detect changes in refractive index or nano mechanical sensor which utilize vibrations to sense mass or force alteration these advancement marks a significant step in sensor technology offering more accuracy more efficiency and more adaptability across numerous applications.

→ From monitoring air quality and detecting air pollutants ~~and~~ to track vital signs in healthcare nano based sensor promise good impact. They enable us to better understanding and interact with our environment, addressing challenges more effectively as nano tech. research progresses. the potential for further innovations in sensor design and functionality grows exponentially. nano material enhanced sensor represent a corner stone of modern tech. ~~so~~ leading to re-shape industries with their unparalleled capabilities these sensors make a future where our ability to detect and response to diverse stimulae is more precise, efficient and accessible than ever before.

3) Application of nanomaterial as a catalyst →

- uses of Nano material in catalysis :-

→ Nano materials are revolutionizing catalysis by enhancing how catalyst interact with substance. they are designed to react with their tiny size and unique surface property make them highly effective in various catalytic process

(a) Hydrogen storage :-

→ ~~E~~ CNT are tiny tube made of carbon atom scientists are interested in them because they can store hydrogen gas research has shown that CNTS can hold more hydrogen than other materials even at room temp. this makes them promising for storing H which is a clean and efficient energy source

(b) water Purification :-

→ Nano particles made of titanium dioxide are used to clean water, titanium dioxide is a special material that can absorb light and create a reaction that can clean the water when light hits titanium dioxide nanoparticle in water it makes electrons move around which helps to breakdown harmful substance in water. Scientists are also looking into using silver nano particle to make water filter that can kill bacteria and keep water clean.

(c) carbon monoxide oxidation in catalytic converter :-

→ catalytic converter are devices in car that cleans up harmful gases from exhausted gases. Nano materials are tiny particles measured in nano meter are used in catalytic converter to clean up CO emission. this converter is

more efficient at reducing pollution from car.

Use of nanomaterial in environment →
→ In environment nanomaterials ~~are~~ ~~for used as~~ ~~fertilizers, water purifier~~

(1) cleaning pollution
(2) Purifying water
(3) Filtering air
(4) Filtering soil
(5) Sensor for monitoring
(6) Generating clean energy
(7) Managing waste
(8) Monitoring the environment
(9) Restoring the Ecosystem

(1) cleaning pollution →
→ NM like graphene oxide can absorb oil in water and titanium dioxident nano particles breakdown pollutants in soil

(2) Purifying air →
→ Nano particles such as CNT are used in water filter to remove contamination and silver NP can kill the bacteria in water purification

(3) Filtering air →
→ Air purifier with NM filter, like nano fiber made from polymer or metal oxide it can trap harmful particles and gases and can improve air quality.

(4) Filtering soil →
→ NP like iron oxide can bind the heavy metal in soil making them easier to remove while nano scale iron can degrade organic pollutants.

(5) Sensor for moni~~to~~ monitoring →
→ NP based sensor such as CNT sensor can detect volatile organic compound and help to provide real time data on environmental

(6) Generating clean energy →
→ Quantum dots enhance the efficiency of solar panel by converting more sunlight into elec. and nano catalyst improve the performance of fuel cell for cleaning the energy product.

(7) Managing waste →
→ NM like Nano zero valent iron breakdown hazardous waste into harmless biproducts.

(8) Monitoring the environment →
→ Smart Nano sensor embedded in environmental monitoring device like nanolab air quality can track pollution level and provide alert to the user

(9) Restoring the Ecosystem →
→ NP like biochar and anchor enhance soil fertility by improving water absorption and nutrient it can promote plant growth and ecosystem restoration.

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